

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Web Programming, Machine Learning & Cloud, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, NLP

SKILLS

• **Languages & DSA:** Python, SQL, Data Structures & Algorithms (Arrays, Hashing, Trees, Graphs, Dynamic Programming), LeetCode

• **Backend & Systems:** FastAPI, REST APIs, Microservices, Distributed Systems, ETL Pipelines, Kafka, PySpark

• **Databases, Cloud & Tools:** PostgreSQL, Neo4j, AWS (S3, EC2, EKS), GCP, Docker, Git

WORK EXPERIENCE

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

- Built and deployed production agentic AI system on Amazon Bedrock and Bedrock Agents using LangChain orchestration, FastAPI, and pgVector, delivering enterprise-scale document intelligence with measurable 64% efficiency gain
- Designed fault-tolerant ETL and async ingestion pipelines with CI/CD automation, structured prompt management, and LLM evaluation hooks for low-latency production-grade retrieval
- Improved system performance by **64%** through optimized indexing strategies and pipeline improvements in production environments
- Developed AI serving layer integrating Bedrock Guardrails and human-in-the-loop review controls for real-time safety enforcement and high-performance LLM inference across enterprise distributed systems

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

- Owned **end-to-end design, development, and integration** of cross-platform **AI backend system** ingesting wearable data for 100+ users, enabling scalable data collection and downstream analytics
- Designed and optimized **PostgreSQL schema, indexing strategies, and distributed query execution plans** for high-volume datasets, improving query performance by 38% in production workloads
- Built production GenAI pipeline using Amazon Bedrock Knowledge Bases and structured prompt engineering to enable RAG-based contextual summarization, improving multi-turn response accuracy by 43%
- Developed async backend workflows with MLflow-based LLM evaluation tracking for real-time, fault-tolerant processing of longitudinal data, ensuring model reliability and observability at scale

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

- Designed scalable GenAI pipelines using LlamaIndex and Hugging Face Transformers in Python, reducing inference latency by 41% through optimized prompt evaluation and embedding retrieval workflows
- Built distributed data ingestion pipelines using **Kafka** and **Neo4j** Streams for real-time, high-throughput processing
- Designed production REST APIs using FastAPI to serve LLM-powered models, implementing prompt versioning, A/B evaluation, CI/CD deployment automation, and monitoring to ensure reliable LLMops delivery across distributed enterprise systems
- Partnered with cross-functional research stakeholders to translate analytical requirements into scalable GenAI data processing workflows across high-volume social media corpora

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

- Built interactive **Power BI** dashboards for **logistics** and **mobility KPIs**, reducing reporting time by **40%** and improving operational visibility
- Processed and optimized large-scale datasets using **Python** and **PySpark** to identify failure and performance issues in distributed systems
- Developed **SQL** and **graph queries** to map dependencies, analyze operations, detect data issues, speeding root-cause analysis by **26%**

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

- Developed and maintained **20+ Power BI** dashboards across Sales, Finance, and Operations for enterprise clients, leveraging **SQL** and **DAX**, enabling KPI tracking and data-driven decisions
- Automated **Excel**-based workflows using **VBA**, implementing validation and scheduling logic, reducing time by **42%** and improving consistency

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

- Led cross-functional team of **6** engineers and designers to deliver **AI-driven product features** for DeepCover game, leveraging feedback and performance insights for feature prioritization
- Tracked **sprint** progress, delivery metrics, and **QA issues** through **JIRA**-based workflows, managing backlog and issue resolution

PROJECTS

Scalable ML System for Customer Risk Prediction

March 2025

UIUC Hackathon

- Built an **end-to-end machine learning system** using Python, Pandas, and LightGBM to forecast customer spending and assess credit risk from large-scale **time-series** transaction data, including data preprocessing, feature engineering, model training, and evaluation
- Developed a **decision framework for credit line adjustments**, incorporating model outputs and risk thresholds to improve prediction precision by 20% and enable risk-aware classification of customers